

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\begin{aligned} \max f(x) &= -2x_1 - x_2 \\ x_1 + x_2 &\geq 5 \\ 2x_1 + x_2 &\geq 8 \\ x_1 + 3x_2 &\geq 6 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Metodo delle due fasi

$$\begin{aligned} \min f(x) &= a_1 + a_2 + a_3 \\ x_1 + x_2 - s_1 + a_1 &= 5 \\ 2x_1 + x_2 - s_2 + a_2 &= 8 \\ x_1 + 3x_2 - s_3 + a_3 &= 6 \\ x_1, x_2, s_1, s_2, s_3, a_1, a_2, a_3 &\geq 0 \end{aligned}$$

Calcolo dei costi

$$c_i = c_N - c_B^T \cdot B_N^{-1}$$

Forma Standard

$$\begin{aligned} \min f(x) &= 2x_1 + x_2 \\ x_1 + x_2 - s_1 &= 5 \\ 2x_1 + x_2 - s_2 &= 8 \\ x_1 + 3x_2 - s_3 &= 6 \\ x_1, x_2, s_1, s_2, s_3 &\geq 0 \end{aligned}$$

$$c_1 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = -4$$

$$c_2 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = -5$$

$$c_3 = 0 - (1; 1; 1) \cdot \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = 1$$

$$c_4 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = 1$$

$$c_5 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 1$$

$$c_6 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_7 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_8 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_9 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 5 \\ 8 \\ 6 \end{pmatrix} = -19$$

Metodo del simplesso

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	1	1	-1	0	0	1	0	0	5
a_2	2	1	0	-1	0	0	1	0	8
a_3	1	3	0	0	-1	0	0	1	6
	-4	-5	1	1	1	0	0	0	-19

Entra x_2 ed esce a_3 :

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	1	1	-1	0	0	1	0	0	5
a_2	2	1	0	-1	0	0	1	0	8
a_3	1	3	0	0	-1	0	0	1	6
	-4	-5	1	1	1	0	0	0	-19

$$R3 \leftarrow R3 \cdot \frac{1}{3}$$

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	1	1	-1	0	0	1	0	0	5
a_2	2	1	0	-1	0	0	1	0	8
a_3	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	-4	-5	1	1	1	0	0	0	-19

$$R1 \leftarrow R1 - R3$$

$$\begin{array}{cccccccc|c} 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 5 \\ \frac{1}{3} & 1 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 2 \\ \hline \frac{2}{3} & 0 & -1 & 0 & \frac{1}{3} & 1 & 0 & -\frac{1}{3} & 3 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	$\frac{2}{3}$	0	-1	0	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	3
a_2	2	1	0	-1	0	0	1	0	8
a_3	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	-4	-5	1	1	1	0	0	0	-19

$$R2 \leftarrow R2 - R3$$

$$\begin{array}{cccccccc|c} 2 & 1 & 0 & -1 & 0 & 0 & 1 & 0 & 8 \\ \frac{1}{3} & 1 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 2 \\ \hline \frac{5}{3} & 0 & 0 & -1 & \frac{1}{3} & 0 & 1 & -\frac{1}{3} & 6 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	$\frac{2}{3}$	0	-1	0	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	3
a_2	$\frac{5}{3}$	0	0	-1	$\frac{1}{3}$	0	1	$-\frac{1}{3}$	6
a_3	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	-4	-5	1	1	1	0	0	0	-19

$$R4 \leftarrow R4 + R3 \cdot 5$$

$$\begin{array}{cccccccc|c} -4 & -5 & 1 & 1 & 1 & 0 & 0 & 0 & -19 \\ \frac{5}{3} & 5 & 0 & 0 & -\frac{5}{3} & 0 & 0 & \frac{5}{3} & 10 \\ \hline -\frac{7}{3} & 0 & 1 & 1 & -\frac{2}{3} & 0 & 0 & \frac{5}{3} & -9 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	$\frac{2}{3}$	0	-1	0	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	3
a_2	$\frac{5}{3}$	0	0	-1	$\frac{1}{3}$	0	1	$-\frac{1}{3}$	6
x_2	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	$-\frac{7}{3}$	0	1	1	$-\frac{2}{3}$	0	0	$\frac{5}{3}$	-9

Entra x_1 ed esce a_2 :

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	$\frac{2}{3}$	0	-1	0	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	3
a_2	$\frac{5}{3}$	0	0	-1	$\frac{1}{3}$	0	1	$-\frac{1}{3}$	6
x_2	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	$-\frac{7}{3}$	0	1	1	$-\frac{2}{3}$	0	0	$\frac{5}{3}$	-9

$$R2 \leftarrow R2 \cdot \frac{3}{5}$$

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	$\frac{2}{3}$	0	-1	0	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	3
a_2	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	$-\frac{7}{3}$	0	1	1	$-\frac{2}{3}$	0	0	$\frac{5}{3}$	-9

$$R1 \leftarrow R1 - R2 \cdot \frac{2}{3}$$

$$\begin{array}{cccccc|c} \frac{2}{3} & 0 & -1 & 0 & \frac{1}{3} & 1 & 0 & -\frac{1}{3} & 3 \\ \frac{2}{3} & 0 & 0 & -\frac{2}{5} & \frac{2}{15} & 0 & \frac{2}{5} & -\frac{2}{15} & \frac{12}{5} \\ \hline 0 & 0 & -1 & \frac{2}{5} & \frac{1}{5} & 1 & -\frac{2}{5} & -\frac{1}{5} & \frac{3}{5} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	-1	$\frac{2}{5}$	$\frac{1}{5}$	1	$-\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
a_2	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	$\frac{1}{3}$	1	0	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	2
	$-\frac{7}{3}$	0	1	1	$-\frac{2}{3}$	0	0	$\frac{5}{3}$	-9

$$R3 \leftarrow R3 - R2 \cdot \frac{1}{3}$$

$$\begin{array}{cccccc|c} \frac{1}{3} & 1 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 2 \\ \frac{1}{3} & 0 & 0 & -\frac{1}{5} & \frac{1}{15} & 0 & \frac{1}{5} & -\frac{1}{15} & \frac{6}{5} \\ \hline 0 & 1 & 0 & \frac{1}{5} & -\frac{2}{5} & 0 & -\frac{1}{5} & \frac{2}{5} & \frac{4}{5} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	-1	$\frac{2}{5}$	$\frac{1}{5}$	1	$-\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
a_2	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	0	1	0	$\frac{1}{5}$	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{4}{5}$
	$-\frac{7}{3}$	0	1	1	$-\frac{2}{3}$	0	0	$\frac{5}{3}$	-9

$$R4 \leftarrow R4 + R2 \cdot \frac{7}{3}$$

$$\begin{array}{cccccc|c} -\frac{7}{3} & 0 & 1 & 1 & -\frac{2}{3} & 0 & 0 & \frac{5}{3} & -9 \\ \frac{7}{3} & 0 & 0 & -\frac{7}{5} & \frac{7}{15} & 0 & \frac{7}{5} & -\frac{7}{15} & \frac{42}{5} \\ \hline 0 & 0 & 1 & -\frac{2}{5} & -\frac{1}{5} & 0 & \frac{7}{5} & \frac{6}{5} & -\frac{3}{5} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	-1	$\frac{2}{5}$	$\frac{1}{5}$	1	$-\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_1	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	0	1	0	$\frac{1}{5}$	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{4}{5}$
	0	0	1	$-\frac{2}{5}$	$-\frac{1}{5}$	0	$\frac{7}{5}$	$\frac{6}{5}$	$-\frac{3}{5}$

Entra s_2 ed esce a_1 :

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	-1	$\frac{2}{5}$	$\frac{1}{5}$	1	$-\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_1	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	0	1	0	$\frac{1}{5}$	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{4}{5}$
	0	0	1	$-\frac{2}{5}$	$-\frac{1}{5}$	0	$\frac{7}{5}$	$\frac{6}{5}$	$-\frac{3}{5}$

$R4 \leftarrow R4 + R1$

$$\begin{array}{cccccc|c}
 0 & 0 & 1 & -\frac{2}{5} & -\frac{1}{5} & 0 & \frac{7}{5} & \frac{6}{5} & -\frac{3}{5} \\
 0 & 0 & -1 & \frac{2}{5} & \frac{1}{5} & 1 & -\frac{2}{5} & -\frac{1}{5} & \frac{3}{5} \\
 \hline
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0
 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	-1	$\frac{2}{5}$	$\frac{1}{5}$	1	$-\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_1	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	0	1	0	$\frac{1}{5}$	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{4}{5}$
	0	0	0	0	0	1	1	1	0

$R1 \leftarrow R1 \cdot \frac{5}{2}$

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{5}{2}$	-1	$-\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	0	1	0	$\frac{1}{5}$	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{4}{5}$
	0	0	0	0	0	1	1	1	0

$R2 \leftarrow R2 + R1 \cdot \frac{3}{5}$

$$\begin{array}{cccccc|c}
 1 & 0 & 0 & -\frac{3}{5} & \frac{1}{5} & 0 & \frac{3}{5} & -\frac{1}{5} & \frac{18}{5} \\
 0 & 0 & -\frac{3}{2} & \frac{3}{5} & \frac{3}{10} & \frac{3}{2} & -\frac{3}{5} & -\frac{3}{10} & \frac{9}{10} \\
 \hline
 1 & 0 & -\frac{3}{2} & 0 & \frac{1}{2} & \frac{3}{2} & 0 & -\frac{1}{2} & \frac{9}{2}
 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{5}{2}$	-1	$-\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	$-\frac{3}{2}$	0	$\frac{1}{2}$	$\frac{3}{2}$	0	$-\frac{1}{2}$	$\frac{9}{2}$
x_2	0	1	0	$\frac{1}{5}$	$-\frac{2}{5}$	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{4}{5}$
	0	0	0	0	0	1	1	1	0

$R3 \leftarrow R3 - R1 \cdot \frac{1}{5}$

$$\begin{array}{cccccc|c}
0 & 1 & 0 & \frac{1}{5} & -\frac{2}{5} & 0 & -\frac{1}{5} & \frac{2}{5} & \frac{4}{5} \\
0 & 0 & -\frac{1}{2} & \frac{1}{5} & \frac{1}{10} & \frac{1}{2} & -\frac{1}{5} & -\frac{1}{10} & \frac{3}{10} \\
\hline
0 & 1 & \frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2}
\end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
s_2	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{5}{2}$	-1	$-\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	$-\frac{3}{2}$	0	$\frac{1}{2}$	$\frac{3}{2}$	0	$-\frac{1}{2}$	$\frac{9}{2}$
x_2	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$
	0	0	0	0	0	1	1	1	0

$z = 0$ si passa alla seconda fase.

Fase II

$$\min f(x) = 2x_1 + x_2$$

$$-\frac{5}{2}s_1 + s_2 + \frac{1}{2}s_3 + \frac{5}{2}a_1 - a_2 - \frac{1}{2}a_3 = \frac{3}{2}$$

$$x_1 - \frac{3}{2}s_1 + \frac{1}{2}s_3 + \frac{3}{2}a_1 - \frac{1}{2}a_3 = \frac{9}{2}$$

$$x_2 + \frac{1}{2}s_1 - \frac{1}{2}s_3 - \frac{1}{2}a_1 + \frac{1}{2}a_3 = \frac{1}{2}$$

Si eliminano le variabili artificiali introdotte nella Fase I

$$\min f(x) = 2x_1 + x_2$$

$$-\frac{5}{2}s_1 + s_2 + \frac{1}{2}s_3 = \frac{3}{2}$$

$$x_1 - \frac{3}{2}s_1 + \frac{1}{2}s_3 = \frac{9}{2}$$

$$x_2 + \frac{1}{2}s_1 - \frac{1}{2}s_3 = \frac{1}{2}$$

Calcolo dei costi della nuova tabella:

$$c_1 = 2 - (0; 2; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_2 = 1 - (0; 2; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_3 = 0 - (0; 2; 1) \cdot \begin{pmatrix} -\frac{5}{2} \\ -\frac{3}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{5}{2}$$

$$c_4 = 0 - (0; 2; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 0 - (0; 2; 1) \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = -\frac{1}{2}$$

$$c_6 = 0 - (0; 2; 1) \cdot \begin{pmatrix} \frac{3}{2} \\ \frac{9}{2} \\ \frac{1}{2} \end{pmatrix} = -\frac{19}{2}$$

Tabella risultante:

	x_1	x_2	s_1	s_2	s_3	b
s_2	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	$-\frac{3}{2}$	0	$\frac{1}{2}$	$\frac{9}{2}$
x_2	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$
	0	0	$\frac{5}{2}$	0	$-\frac{1}{2}$	$-\frac{19}{2}$

Entra s_3 ed esce s_2 :

	x_1	x_2	s_1	s_2	s_3	b
s_2	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	$-\frac{3}{2}$	0	$\frac{1}{2}$	$\frac{9}{2}$
x_2	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$
	0	0	$\frac{5}{2}$	0	$-\frac{1}{2}$	$-\frac{19}{2}$

$R3 \leftarrow R3 + R1$

$$\begin{array}{cccc|c} 0 & 1 & \frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{5}{2} & 1 & \frac{1}{2} & \frac{3}{2} \\ \hline 0 & 1 & -2 & 1 & 0 & 2 \end{array}$$

$\downarrow$

	x_1	x_2	s_1	s_2	s_3	b
s_2	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	$-\frac{3}{2}$	0	$\frac{1}{2}$	$\frac{9}{2}$
x_2	0	1	-2	1	0	2
	0	0	$\frac{5}{2}$	0	$-\frac{1}{2}$	$-\frac{19}{2}$

$$R2 \leftarrow R2 - R1$$

$$\begin{array}{cccc|c} 1 & 0 & -\frac{3}{2} & 0 & \frac{1}{2} & \frac{9}{2} \\ 0 & 0 & -\frac{5}{2} & 1 & \frac{1}{2} & \frac{3}{2} \\ \hline 1 & 0 & 1 & -1 & 0 & 3 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
s_2	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	1	-1	0	3
x_2	0	1	-2	1	0	2
	0	0	$\frac{5}{2}$	0	$-\frac{1}{2}$	$-\frac{19}{2}$

$$R4 \leftarrow R4 + R1$$

$$\begin{array}{cccc|c} 0 & 0 & \frac{5}{2} & 0 & -\frac{1}{2} & -\frac{19}{2} \\ 0 & 0 & -\frac{5}{2} & 1 & \frac{1}{2} & \frac{3}{2} \\ \hline 0 & 0 & 0 & 1 & 0 & 8 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
s_2	0	0	$-\frac{5}{2}$	1	$\frac{1}{2}$	$\frac{3}{2}$
x_1	1	0	1	-1	0	3
x_2	0	1	-2	1	0	2
	0	0	0	1	0	8

$$R1 \leftarrow R1 \cdot 2$$

	x_1	x_2	s_1	s_2	s_3	b
s_3	0	0	-5	2	1	3
x_1	1	0	1	-1	0	3
x_2	0	1	-2	1	0	2
	0	0	0	1	0	8

$$x_1 = 3$$

$$x_2 = 2$$

$$s_3 = 3$$

La variabile di slack s_1 ha costo ridotto 0 il simplesso può essere reiterato ancora, ciò significa che esistono infiniti punti di ottimo.